Friedrich-Alexander-Universität
Technische Fakultät

Towards LLM Brainscore: What Probing experiments reveal about the inner workings of LLMs

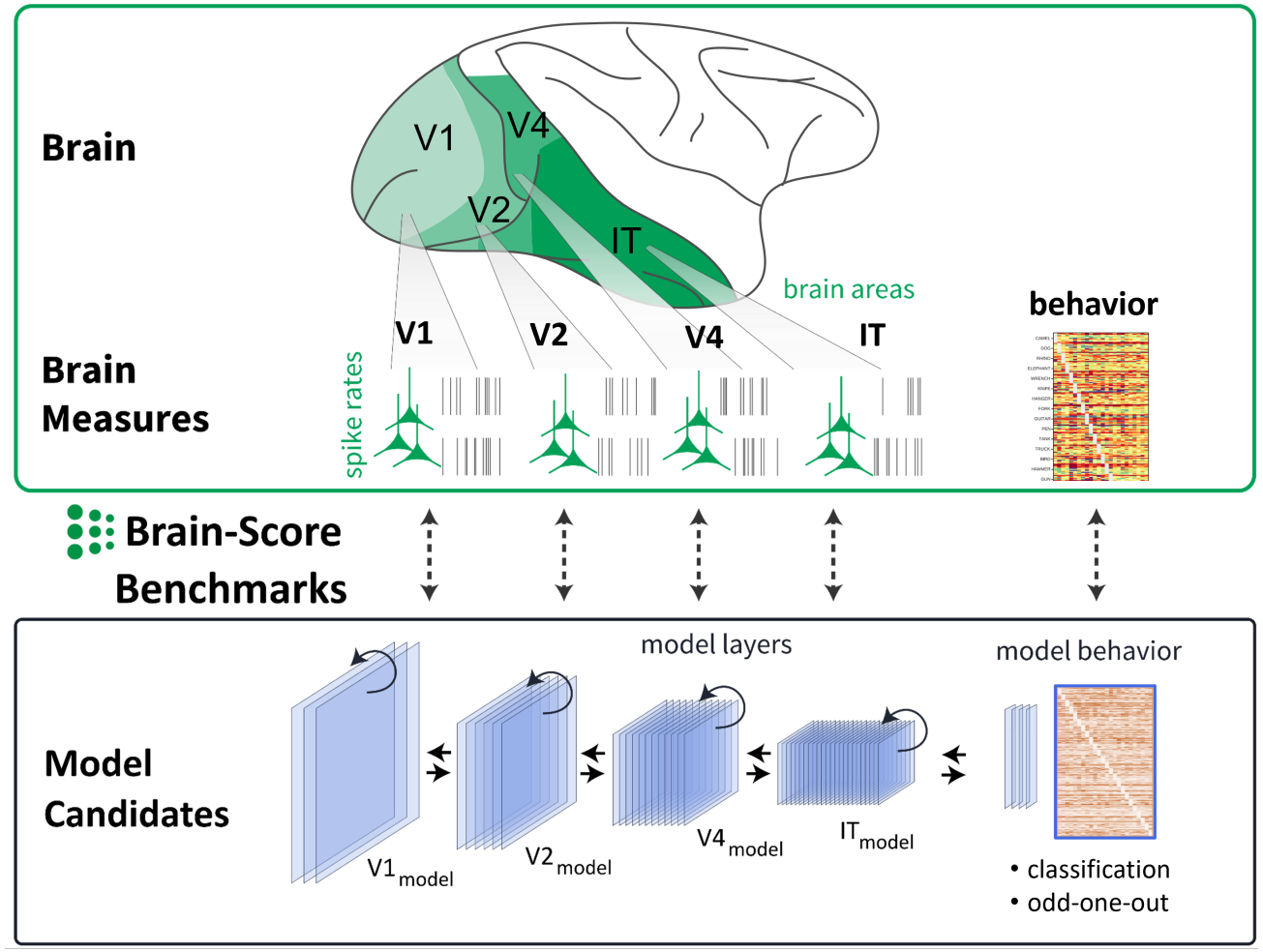
Julia Isabel Hagen M.Sc. Artificial Intelligence

Research

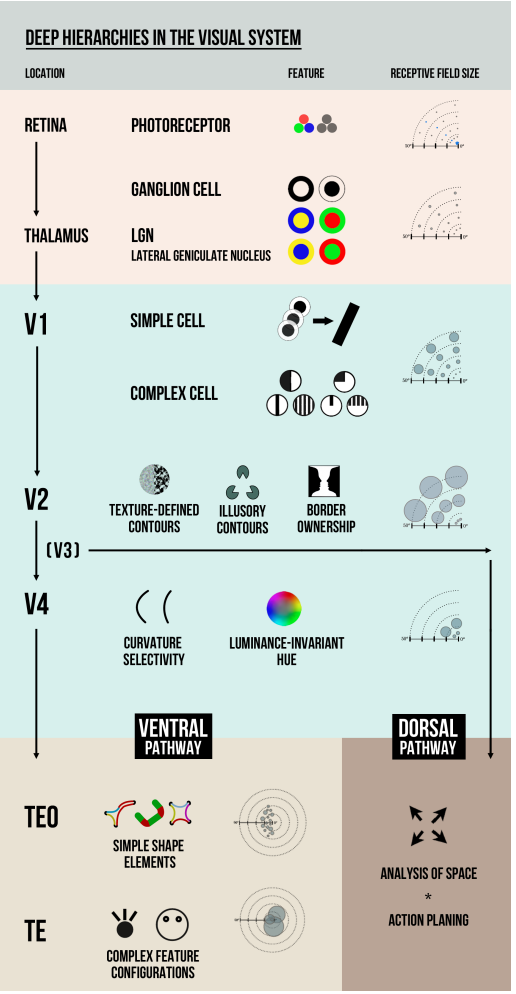
FAU, Computational Cognitive Neuroscience Lab	Integrated Neuroscience & AI Research: Towards LLM BrainScore & More
ZHAW, Institute for Data Analysis and Process Design	Inside Vision Transformers: Interestingness Decoded
FAU, Nsquared Lab	Real-Time EMG Processing for Prosthetic Control
BMW, Predevelopment	AI Software Engineer & Researcher Blackbox E2E GenAI Testing
Vencortex	LLMs for TimeSeries & Statistics Based Tokenization Protocols

Vision

Why is it the same & why is it different?

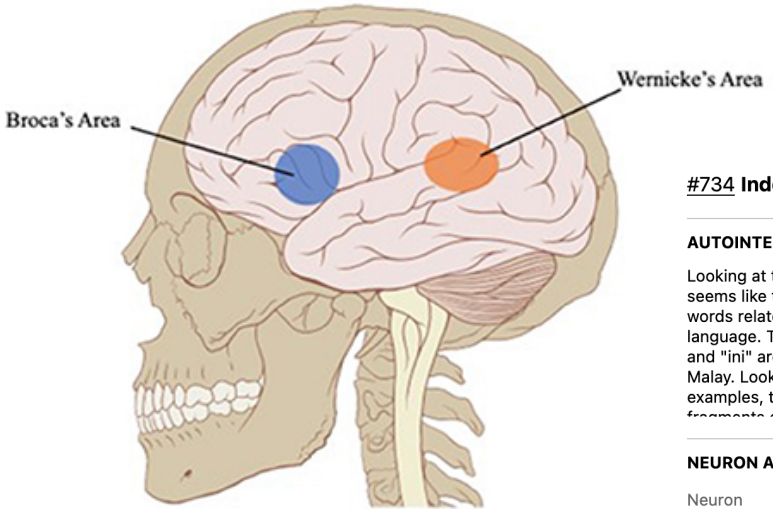


2024 Brain-Score Competition." Brain-Score, n.d. Web. 22 June 2025. https://www.brain-score.org/competition/

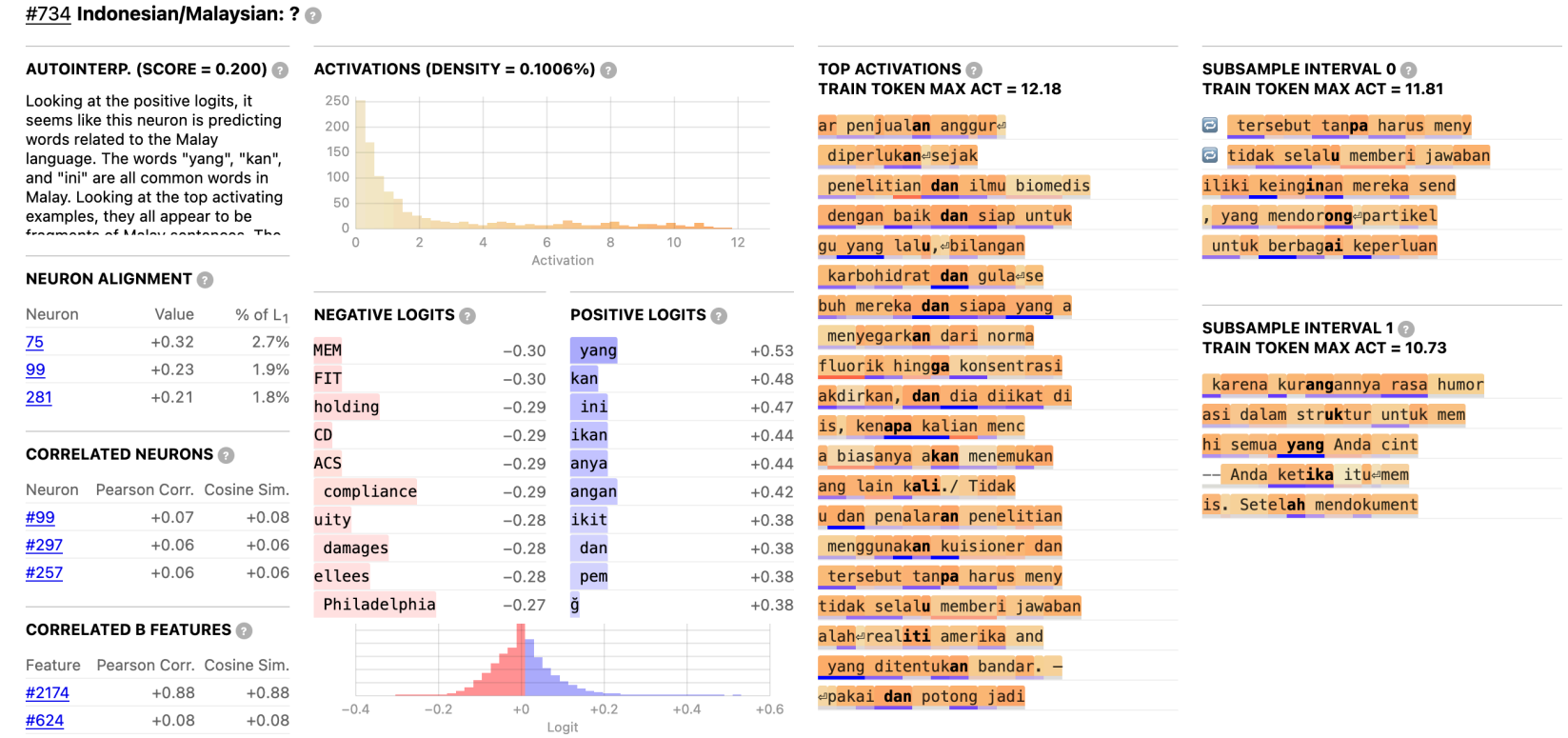


Kanders, Karlis. "Deep hierarchies in the visual system." Wikimedia Commons, 7 Sept. 2014. CC BY-SA 4.0. https://commons.wikimedia.org/wiki/File:Wiki_hierarchies.png

Motivation



Lynch, P. J. (2014). Skull and brain normal human [PNG]. Wikimedia Commons. CC BY 2.5. Available at https://commons.wikimedia.org/wiki/File:Skull_and_brain_normal_human.svg



Transformer-Circuits. (2023). Monosemantic Features Visualization: A1 – Dead [Visualization]. Retrieved June 22, 2025, from https://transformer-circuits.pub/2023/monosemantic-features/vis/a1.html#dead

Language Processing Task

Bridging between modalities

Language Processing Task (LPT)

Hypothesis:

- **Stimulus– response?**
- **Behavioural vs. neural?**

Challenge:

Condensing LPT into Data & Target

Problem: Many ML Problems subsume one LPT

Idea: Aggregate over multiple tasks designed for ”provoking” semantic responses in LLMs

Language Processing Task: Semantics

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Task

Hypothesis

Data

Semantics

Semantic processing (Neural)

- **Homonyms** in **semantic proximity** after tokenization
- **Differentiation with increasing context**

Synthetic Data Generation

Word in Context: Homonyms

- Eliminates Syntactic influence to illustrate only Semantic influence on activations
- Example: “river bank” vs. ”financial bank”

Namesake Differentiation

Expert (Fine-tuning) Differentiation

Increased within-domain differentiation for fine-tuned expert models

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Synthetic Data Generation

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How? **Methods…**

Where? **Query Target**

What? **Interpretation**

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Methods: How to make them interpretable?

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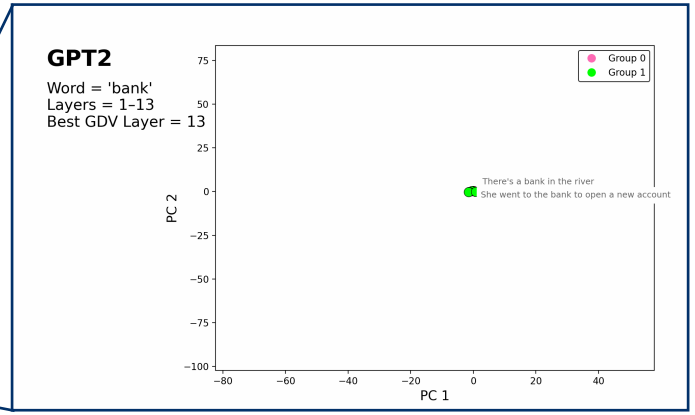

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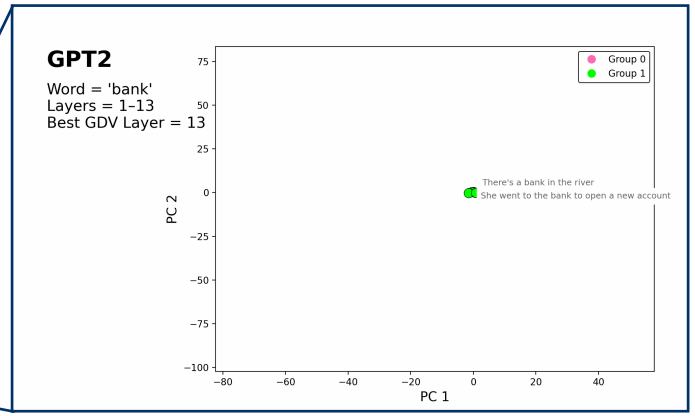

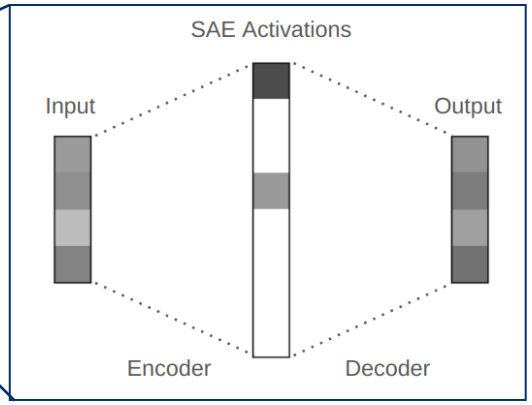

Karvonen, A. (2024, June 11). An Intuitive Explanation of Sparse Autoencoders for LLM Interpretability [Web page]. Adam Karvonen's Machine Learning. Available at https://adamkarvonen.github.io/machine_learning/2024/06/11/sae-intuitions.html

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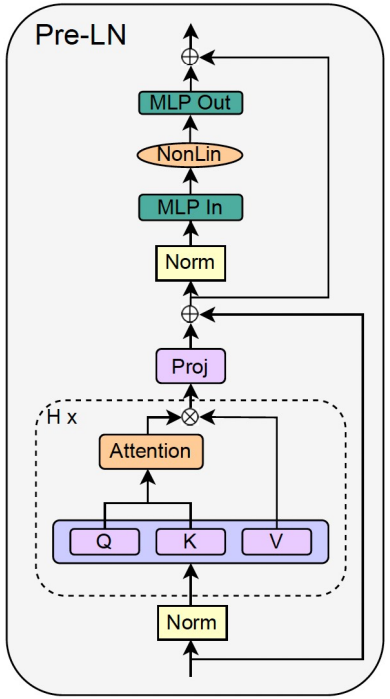

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., & Polosukhin, I. (2017). Classic transformer block [PNG]. arXiv. Non-exclusive distribution license. Available at https://arxiv.org/abs/1706.03762

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Control: What can we learn from that about function?

- Deterministic control of reaction or function through deliberately **clamping** i.e. concept neurons 🔗 Anthropic
- **Lesioning/ablation:** can we disable a concept or function deterministically by removing the component identified as responsible?

Within LLM comparison: track embedding ”performance” across layer of single LLM to identify semantically active components

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Across (multiple) LLMs comparison:

Hypothesis:
LLM layers, which fulfill a **specialized LPT in one LLM have a corresponding layer in another LLM**

Methods:
- Representational Similarity Analysis (RSA)
- Statistical Tests on Probing Classifier Results

Validation:
- C.f. Control
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Results (preliminary):

Within-LLM

- **Homonyms** appear in **semantic proximity** after tokenization confirming hypothesis
- **Statistically significant differences between within-LLM layer on-task performance**

Across-LLM

- **Same LPT goal through different means**: semantically active layers appear in different orders: e.g. semantics appear early in BERT and late in GPTs (bottom-up vs. top-down processing?)
- **Monosemantic neurons scattered across layers**: no „semantic layer“ ?
- **High dimensionality of attention suggests multi-task layers**

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- **High semantic content right after tokenization**: Mental Lexicon in Neuroscience ?
- **Is the parallelization through multitask layers actually closer to biological processing?**

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Biological vs. Artificial Intelligence

2 Setups:
- **Similarity Measure c.f. BrainScore**
- **Functional/ Hypothesis driven Experimentation**

Hypothesis:
Components/ Concepts/Functions are shared across systems with the same capabilities

Methods:
- Same as within LLM on Biological Neural Data
- Comparative Methods

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Vielen Dank
für Ihre Aufmerksamkeit!